11 Organic chemistry

1. Functional group chemistry (alkane, alkene, alchanol, carboxylic acid)

2012-3.

The following are general formulae of organic compounds **A** and **B**.

A. $C_nH_{2n+1}COOH$

B. $C_nH_{2n+1}OH$

a. To which family does compound **B** belong? (1mark)

b. Mention any three properties of compound **A**. (3marks)

c. State any three uses of compound **B**. (3marks)

d. Mention the products formed when compounds **A** and **B** react. (2marks)

e. Work out molecular formula of compound **A** if **n** is **5**. (3 marks)

f. Describe how compounds **A** could be distinguished from compound **B**. (5 marks)

2011-5.

b. Mention any two properties of alkanols. (2 marks)

c. Ethanol (CH_3CH_2OH) changes to ethanoic acid (CH_3COOH) in the presence of atmospheric oxygen (0_2).

(i) What is the function of atmospheric oxygen in the reaction? (1 mark)

(ii) Write a balanced chemical equation for the reaction. (3 marks)

2010-4.

b.

(i) Give three properties of carboxylic acids. (3 marks)

(ii) Mention any two natural sources of carboxylic acids. (2 marks)

d.

Ethanol (C_2H_5OH) reacts with ethanoic acid (CH_3COOH) according to the following chemical equation.

$$CH_3COOH + C_2H_5OH \underset{H_2SO_4}{\overset{\text{Concentrated}}{\rightleftharpoons}} \underline{\qquad\qquad\qquad} + H_2O$$

(i) Complete the equation. (1 mark)

(ii) Name the process in which ethanol reacts with ethnoic acid. (1 mark)

2010-8.

c. Explain why propanoic acid (C_2H_5COOH) conducts electricity when dissolved in water while propanol (C_3H_7OH) does not.

2009-5.

a. State any three uses of ethanoic acid. (3marks)

b. Why is the ethanoic acid regarded as a weak electrolyte? (1mark)

c. Write down the ionization equation of ethanoic acid (CH_3COOH) in water(H_2O). (4marks)

d. Why does sodium metal react with ethanol in the same way as it does with water? (1mark)

f.

(i) Write down the general formula for carboxylic acids. (1mark)

(ii) What is the formula and name of the smallest carboxylic acid? (2marks)

(iii) How would the boiling point of butane compare with that of a carboxylic acid of similar size? (1mark)

(iv) Explain your answer to 5.f.(iii). (3marks)

2008-5.

c. **Figure 5** is a diagram of an experimental set up.

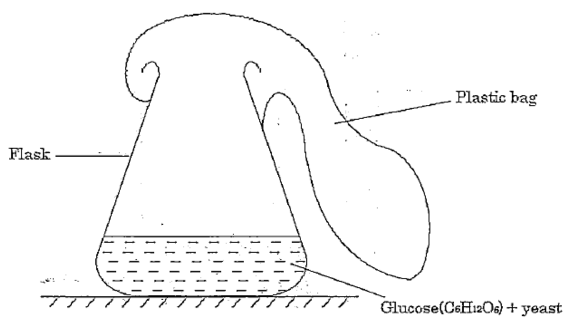


Figure 5

(i) Name the process that could occur in the flask. (1 mark)

(ii) Write down a balanced equation of the process named in 5.c.(i). (3 marks)

e. **Table 2** shows boiling points and solubility of some alkanols in water.

Name	Boiling point (°C)	Solubility
Methanol	65	Most soluble
Ethanol	78	Soluble
Propanol	97	Partially soluble
Butanol	117	Insoluble

(i) Explain why the boiling points of alkanols increase from methanol to butanol. (3 marks)

(ii) Explain why ethanol is more soluble in water than propanol. (8 marks)

f.

(i) Complete the following equation to show the reaction between methane and chlorine.

$CH_4 + Cl_2 \rightarrow$ ________________ + __________________ (2 marks)

(ii) Name this type of reaction. (1 mark)

(in) Give any one use of alkanes. (1 mark)

2008-7.

a. Describe an experiment that can be done to distinguish octane from octene. (5 marks)

2007-1.

b. the following are formulae of some organic compounds:

A. $CH_3CH_2CH_2CH_2OH$

B. $CH_3CH_2CH_3$

C. $CH_3CH_2CH_2COOH$

D. $CH_3CH_2CH{=}CH_2$

E. $CH_3CH_2CH_2CH_3$

(i) Identify one compound which is an alkanol. (1mark)

(ii) Which compounds belong to the same homologous series? (1mark)

(iii) Explain why a solution of compound **C** conducts electricity. (2marks)

(iv) Draw a full structure of compound **D**. (1mark)

(v) Name compound **D**. (1mark)

2007-7.

a. Draw full structures of ethanol (C_2H_5OH) and water (H_2O). (2marks)

b. Explain the difference in boiling points between ethanol and water. (5marks)

c. With the help of a labeled diagram, describe an experiment that can be done to separate a mixture of ethanol and water. (8marks)

2006-1.

a. Give below is the general formulae of some homologous series represented by letters **P**, **Q**, **R**, and **S**.

P: C_nH_{2n}

Q: C_nH_{2n+2}

R: $C_nH_{2n+1}OH$

S: $C_nH_{2n+1}COOH$

(i) Name the homologous series represented by letters **Q** and **S**. (1mark each)

(ii) Which general formulae represent hydrocarbons? (2marks)

(iii) Draw the structure of a compound with three carbon atoms in homologous series **P**. (2marks)

(iv) Name the compound drawn in 1.a. (iii). (1mark)

(v) Explain how a compound of homologous series **Q** could be distinguished from a compound of homologous series **R**. (1mark)

c.

Ethene (C_2H_4) reacts with bromine (Br_2) in an addition reaction.

(i) Draw the structure of the product formed. (1mark)

(ii) Name the product formed in 1.c. (i). (1mark)

(iii) Why are addition reactions important in industries? Give two reasons. (2marks)

2005-5.

a. **Figure 5** is a diagram showing how ethanol and tannic acid are produced.

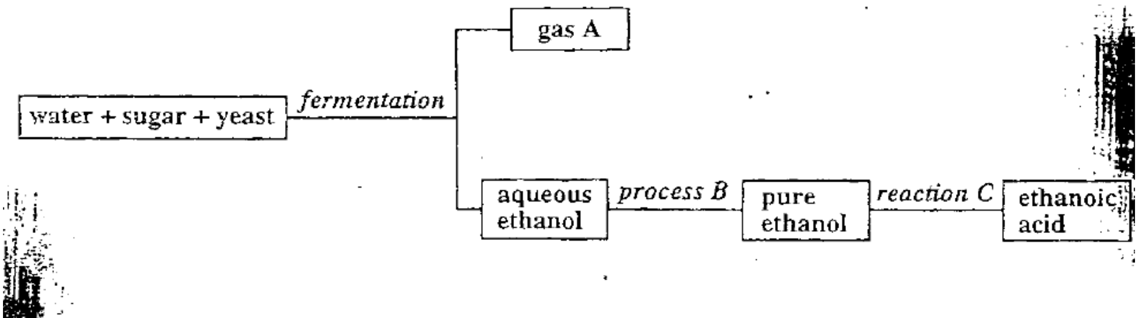


Figure 5

(i) Give the names of

(1) Gas A; (1mark)

(2) Process B; (1mark)

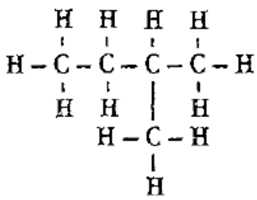
(3) Reaction C. (1mark)

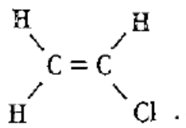
(ii) Name the substance that is used in reaction C. (1mark)

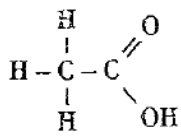
(iii) What is the function of the substance in a (ii)? (1mark)

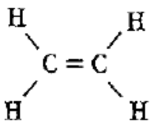
2005-5.

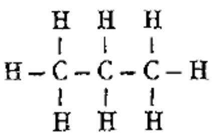
b. the following is structures of some organic compounds.

A. 

B. 

C. 

D. 

E. 

(i) Name compound **A.** (1mark)

(ii) Which compound is soluble in water? Give a reason. (2marks)

(iii) Write letters representing any three compounds that would not react with potassium, a group 1 metal element. (3marks)

(iv) Which one of the two compounds **A** and **E** would have a lower boiling point? Give a reason. (2marks)

2004-3.

d. **Table 4** shows molecular formulae and boiling points of some compounds.

Table 4

COMPOUND	MOLECULAR FORMULA	BOILING POINT (°C)
A	C_2H_4	–104
B	C_2H_5OH	79
C	CH_3COOH	118
D	H_2O	100
E	C_2H_6	–89

(i) Which compounds in the table are hydrocarbons? (2 marks)

(ii) Which compounds in the table are soluble in water? (2 marks)

(iii) Which compounds in the table are gases at room temperature? (2 marks)

(iv) Explain why the boiling point of compound **D** is higher than the boiling point of compound **E**. (4marks)

(v) Describe a test which can be done to distinguish the compounds **C** and **D**. (4 marks)

2003-5.

a.

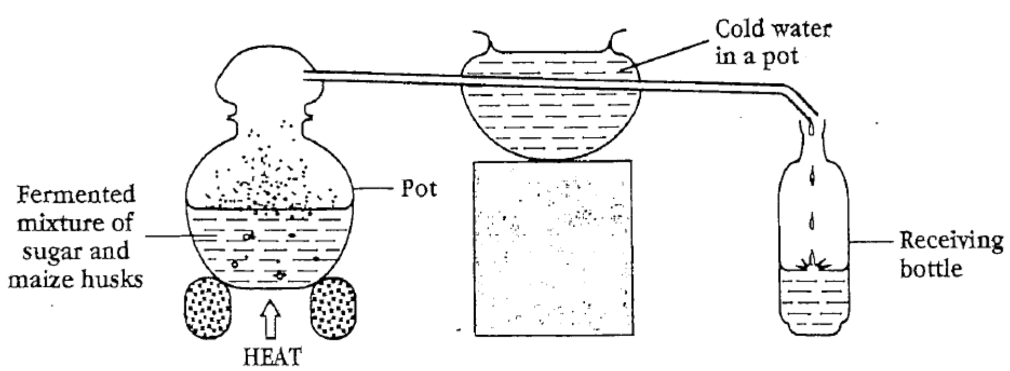
(i) Name the compound $C_7H_{15}OH$ (1mark)

(ii) What is the general formula of the compound in 5.a(i)? (1 mark)

(iii) Draw the structure of the compound $C_7H_{15}OH$ (1 mark)

b.

Figure4 shows one indigenous way of preparing alcohol.



(i) Name the process illustrated in **Figure4**. (1 mark)

(ii) Name the alcohol collected in the receiving bottle (1 mark)

(iii) The alcohol collected in figure4 is produced by fermentation.

(1) Define "fermentation". (2 mark)

(2) Write a word equation for the fermentation of sugar. (3marks)

c. Ethanoic acid (CH_3COOH) reacts with ethanol (C_2H_50H) according to the following equation.

$$CH_3COOH(l) + C_2H_5OH(l) \xrightarrow[\text{Heat}]{\text{Conc. } H_2SO_4} CH_3COOC_2H_5(l) + H_2O(l)$$

(i) What is the name of this reaction? (1mark)

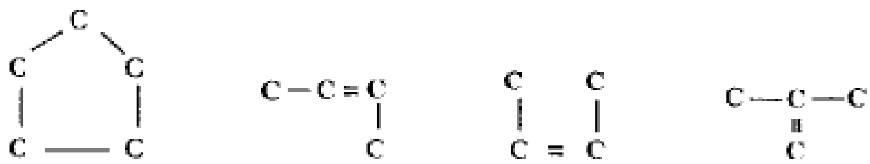
(ii) Name the two products of this reaction. (2marks)

(iii) Give one use of $CH_3COOC_2H_5$. (1mark)

2. Isomerism

2011-5.

d. **Figure 3** shows structures of some organic compounds **P**, **Q**, **R** and **S**.



P Q R S

Figure 3

(i) Name the organic compounds labeled **P** and **Q**. (1 mark each)

(ii) Identify the structure which is an isomer of compound **Q**. (1 mark)

(iii) Give a reason for the answer in 5.d.(ii). (1 mark)

(iv) Explain how compound **P** could be distinguished from compound **R** using bromine solution. (3 marks)

2010-4.

c.

(i) Define “isomers”. (2 marks)

(ii) Draw structural formulae for the four isomers of butanol (C_2H_9OH). (4 marks)

(iii) Write down the condensed formula of pentane. (2 marks)

2008-5.

a. Define “isomers”.

b.

(i) Draw structural formulas for the two isomers of butane (C_4H_{10}).

(ii) Name the isomers draw in 5.b.(i). (1 mark each)

2006-1.

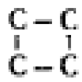
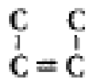
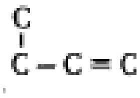
b.

(i) Write down all structural isomers of pentane. (3marks)

(ii) Name the isomers in 1.b. (i). (3marks)

2004-3.

c. The following are structural formulae of four molecules with the molecular formula C_4H_8.

1  2  3  4 

(i) Name the molecules **1** and **2**. (1 mark)

(ii) Which two structures are conformations of each other? (1 mark)

2003-5.

d. Draw and name all the isomers of pentane(C_5H_{12}). (6marks)

<u>3. Polymerisation</u>

2011-5.

a.

(i) Give any two properties of polymers. (2 marks)

(ii) Explain how “condensation polymerization" occurs. (2 marks)

2010-4.

a.

(i) What are “polymers”? (1 mark)

(ii) Mention any two uses of polythene. (2 marks)

(iii) Give any three properties of plastics. (3 marks)

2010-8.

d. Explain how polythene is formed. (3 marks)

2009-5.

g. State any ‘three ways of managing plastic wastes. (3marks)

2009-7.

a. Explain any three characteristics of thermoplastics. (6marks)

b. Explain any two advantages of recycling organic compounds. (4marks)

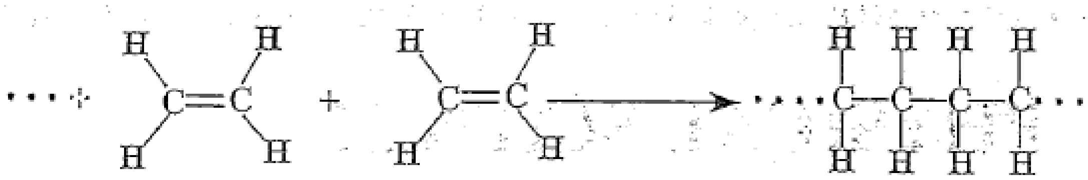
c. Explain why thermosetting plastics can be heated and moulded only once. (5marks)

2008-5.

d. State any two disadvantages of synthetic polymers. (2 marks)

2007-1.

a. Polymerization of Ethene can be represented by the following equation:



(i) Name the polymerization represented by the equation. (1mark)

(ii) Describe how the polymer is formed from ethene molecules. (3marks)

(iii) Give two examples of artificial polymers. (2marks)

c.

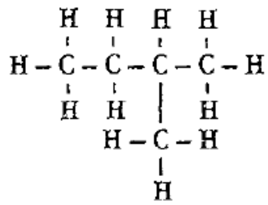
(i) Give three differences between thermosetting and thermoplastic polymers. (3marks)

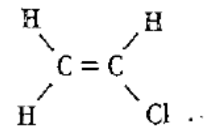
(ii) State two ways of disposing of plastic waste to avoid pollution. (2marks)

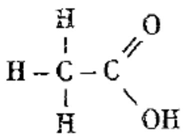
(iii) Give three advantages of plastic materials over metallic materials. (3marks)

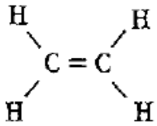
2005-5.

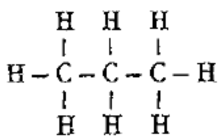
b. the following is structures of some organic compounds.

A. 

B. 

C. 

D. 

E. 

(v) Compound B is a monomer. Write an equation to show its polymerization. (2marks)

(vi) Give the name of the kind of polymerization in b(v). (1mark)

(vii) Give one use of the substance formed in the polymerization of compound. (1mark)

(x) Write the other isomers of substance A. (2marks)

c. Give two advantages of thermoplastics. (2marks)

2004-3.

a. State one use of each of the following polymers.

(i) plastic

(ii) carbohydrate (2 marks)

b. State any two ways of disposing of plastics to avoid polluting the environment. (2 marks)